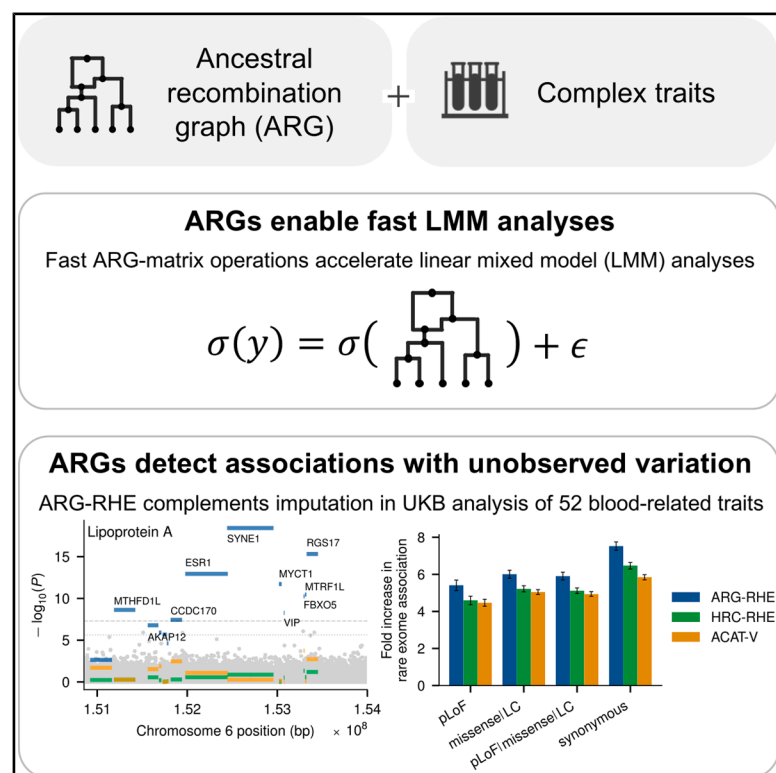


Leveraging ancestral recombination graphs for scalable mixed-model analysis of complex traits

Graphical abstract



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In brief

Zhu et al. present scalable approaches for leveraging genealogical graphs to study complex traits. They introduce ARG-RHE, a method for heritability estimation and gene-based association testing that, when applied to inferred genealogies from 337,464 UK Biobank participants, identifies associations in 52 blood traits that complement genotype imputation.

Highlights

- Sublinear-time ARG-based matrix operations enable efficient linear mixed-model analyses
- ARG-RHE estimates heritability and tests for gene-based association on inferred ARGs
- ARG-RHE complements genotype imputation in detecting rare-variant associations



Technology

Leveraging ancestral recombination graphs for scalable mixed-model analysis of complex traits

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SUMMARY

Recent algorithmic advances have enabled the inference of genome-wide ancestral recombination graphs (ARGs) from large genomic cohorts, providing detailed models of genealogical relatedness along the genome. These inferred ARGs can complement genotype imputation by capturing the effects of unobserved variants, but their use in large-scale linear mixed-model analyses has been computationally prohibitive. Here, we develop methods that leverage the ARG to perform genotype-matrix multiplications in sublinear time and implement scalable randomized algorithms for mixed-model analyses. We introduce ARG-RHE, a randomized Haseman-Elston approach for estimating narrow-sense heritability and performing region-based association testing using ARGs, enabling parallel analysis of multiple quantitative traits. Through extensive simulations, we demonstrate the computational efficiency and statistical power of this approach. Applied to 21,159 genes and 52 blood traits in 337,464 UK Biobank participants, ARG-RHE identifies 8% more gene-trait associations than imputation alone, demonstrating that genome-wide genealogies may be leveraged to complement genotype imputation in complex trait analyses.

INTRODUCTION

Modern biobank initiatives have led to the collection of growing volumes of genomic and phenotypic information.^{1–5} These datasets have been utilized in various applications, including the detection of phenotypic associations,^{6,7} the study of complex trait architectures,^{8,9} and the development of polygenic scores.¹⁰ A key step in these analyses is to study the patterns of genetic and genealogical relationships among the analyzed samples, which provide insights into phenotypic and environmental variation. A wide range of approaches has been developed to analyze details of these relationships, ranging from methods that use genomic data to detect shared recent ancestry^{11–16} to approaches that reconstruct genome-wide genealogical relationships using an ancestral recombination graph (ARG).^{17–20}

Inferred genome-wide genealogies have predominantly been used to facilitate analyses in population genetics^{17–19,21} but have also recently been used for the study of complex traits.^{20,22,23}

One approach is to use an ARG inferred from a subset of common variants to detect the presence of unobserved genomic variation, which can then be tested for association with a phenotype of interest.^{20,24} The unobserved variation inferred from the ARG can also be used to construct a genetic relatedness matrix (GRM), which in turn can be used to estimate heritability, perform polygenic prediction, or detect associations within the framework of linear mixed models (LMMs). In this work, we adopt the terminology of Zhang et al.,²⁰ where ARG-derived GRMs were introduced for LMM-based complex trait analyses and referred to as ARG-GRMs. A contemporary approach²⁵ independently described a similar model for population genetic analyses, referred to as eGRM. Both models build on related work on the genealogical interpretation of principal-component analysis.²⁶

When used to estimate heritability and perform other mixed-model analyses, ARG-GRMs can improve performance by capturing the contribution of variants that would otherwise remain unobserved.²⁰ This strategy can also be utilized for



variance component association testing²³ by constructing the GRM for a specific region of interest and assessing whether heritability significantly deviates from zero. By reducing the number of tests performed and pooling potentially heterogeneous effects within a region, as is often done in the analysis of rare variants,^{27–30} this ARG-based variance component testing approach may lead to increased association power. However, estimating global or local heritability using the approaches of Zhang et al.²⁰ and Link et al.²³ requires generating and manipulating ARG-GRMs that contain $n \times n$ entries for n studied individuals, which becomes computationally infeasible in datasets comprising more than a few thousand individuals.

Here, we develop computationally efficient methods that leverage graph traversal and randomized algorithms to improve the scalability of several ARG-based analyses.²⁰ We develop algorithms that perform general matrix multiplication within the ARG in sublinear time, enabling large-scale mixed-model analysis and other downstream applications. We then focus on variance component analysis and present ARG-RHE, a randomized Haseman-Elston regression algorithm for estimating heritability and performing region-based association testing using the ARG. ARG-RHE is highly scalable, making it applicable to ARGs inferred from large biobank datasets, and enables modeling of phenotypic effects from variants that are not genotyped or reliably imputed using sequencing reference panels. It combines ARG-based matrix multiplication with two existing methodological frameworks: a randomized algorithm for moment-based estimation of narrow-sense heritability (RHE-mc^{31,32}) and moment-based approximations for computing tail probabilities in significance testing (FastSKAT^{28,33}). We apply ARG-RHE to detect associations between 52 quantitative blood-related traits and an ARG inferred from SNP array data for 337,464 individuals in the UK Biobank (UKBB²). We find that the associations detected using ARG-RHE are complementary to those identified by testing ~54 million variants imputed from ~65,000 sequenced haplotypes in the Haplotype Reference Consortium (HRC³⁴) panel and are enriched for rare-variant associations detected using exome sequencing data.³⁵

DESIGN

Model formulation

LMMs provide a standard framework for analyzing complex traits, supporting key analyses such as heritability estimation, polygenic prediction, and mixed-model association.^{36,37} We consider the model $y = W\gamma + X\beta + e$, where y is the phenotype vector for n samples, W is a set of covariates (e.g., age and sex) with fixed effects γ , X is the $n \times m$ genotype matrix for m variants, β denotes random SNP effect sizes, and e denotes residual noise. The random components $g = X\beta$ and e represent the additive genetic and environmental effects, with $\beta \sim \mathcal{N}\left(0, \frac{\sigma_g^2}{m}I_m\right)$ and $e \sim \mathcal{N}(0, \sigma_e^2 I_n)$. This implies that $g \sim \mathcal{N}(0, \sigma_g^2 R)$, where $R = XX^T/m$ is the $n \times n$ GRM. In practice, X is often restricted to variants that can be reliably genotyped or imputed, so causal variants may be poorly tagged, leading to

missed associations, downward-biased heritability estimates, and reduced prediction accuracy. Genomic variants, however, arise stochastically during transmission between relatives and can be viewed as random samples from the ARG, which encodes genealogical relationships along the genome. Zhang et al.²⁰ therefore proposed leveraging the ARG to capture the contribution of unobserved variants within the LMM framework, estimating the GRM using its expectation conditional on observing the ARG. In this construction, the entry R_{ij} of such an ARG-derived GRM (ARG-GRM) is proportional to the genealogical distance between individuals i and j , reflecting the expected number of sites at which their sequences differ, with additional modeling for the relationship between allele frequency and effect sizes (see supplementary note 3 of Zhang et al.²⁰). This approach was shown to improve downstream LMM analyses in simulation²⁰ but required explicitly constructing $n \times n$ ARG-GRMs using the `arg-needle-lib` library and analyzing them with `GCTA`,³⁶ resulting in computational costs that limit scalability to biobank-scale data.

Proposed methodology

In this work, we developed algorithmic advances to overcome these computational limitations by exploiting the ARG to accelerate matrix multiplications central to LMMs and other genomic analyses and by integrating this approach into efficient LMM formulations. Our approach for ARG-based matrix multiplication (hereafter, ARG-matrix multiplication) uses graph traversal with memoization to propagate information along ARG nodes and edges. This enables matrix-vector products to be computed without explicitly forming the genotype matrix, with computational cost that is often sublinear in sample size depending on the extent of linkage disequilibrium (LD). ARG-matrix multiplication can be applied to several key numerical routines that rely on matrix-vector products, including randomized singular value decomposition (SVD)^{38,39} for eigen-decomposition of the GRM, Hutchinson's trace estimation,⁴⁰ and conjugate gradients for solving linear systems of the form $Vx = y$, where $V = \sigma_g^2 R + \sigma_e^2 I_n$ is the phenotype covariance matrix under the LMM. These routines can accelerate a range of ARG-based analyses, of which we provide some examples.

Building on this framework, we develop ARG-RHE, a method for ARG-based heritability estimation and variance component association testing. ARG-RHE implements moment-based heritability estimation via RHE regression,^{31,32} avoiding explicit GRM construction by relying on ARG-matrix multiplication. It uses Hutchinson's estimator to approximate traces such as $\text{Tr}R^2$ and randomized SVD to approximate leading eigenvalues of R to evaluate the tail probabilities of the quadratic form test statistic $y^T R y$.³³ We evaluate the scalability, calibration, and power of ARG-RHE in simulations and apply it to an ARG inferred from 337,464 UKBB individuals, where we perform gene-based variance component association tests across 52 blood traits. These analyses demonstrate that efficient ARG-based LMM methods can scale to biobank-sized cohorts and complement imputation-based approaches by recovering association signals that may otherwise remain undetected.

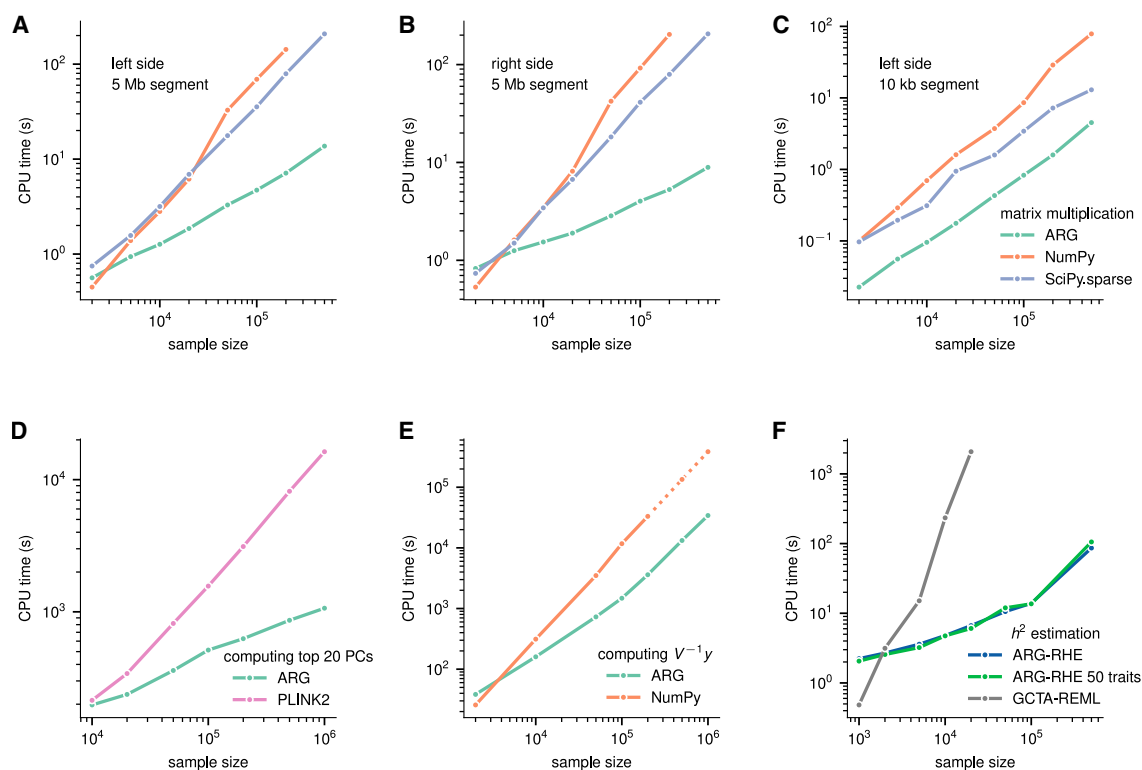


Figure 1. Scalability of ARG matrix multiplication algorithm

(A–C) CPU time taken to calculate genotype matrix-matrix product using various methods, based on the average runtime from 10 replicates. In (A) and (C), left-side multiplication computes UX , and in (B), right-side multiplication computes XU , where U is a random input matrix with one dimension implied by the $n \times p$ genotype X , and the other being 100. In (A) and (B), ARGs are simulated under the GBR demographic model and cover a 5-Mb region, with mutations resampled on the ARG at a rate of 10^{-8} . In (C), ARGs are simulated under the GBR demographic model and cover a 10-kb region, with mutations resampled on the ARG at a rate of 10^{-6} .

(D) CPU time taken to calculate top 20 principal components using various methods. ARGs are simulated under the GBR demographic model and cover a 30-Mb region. Mutations are resampled on the ARG at a rate of 10^{-8} and filtered to have a MAF of at least 1%. 4 power iterations are used in the ARG-based randomized PCA scheme.

(E) CPU time required to compute $V^{-1}y$, used to obtain the BLUP or to residualize the phenotype y for LMM association testing, where $V = \frac{\sigma_a^2}{m}XX^T + \sigma_e^2I_n$. This is computed using conjugate gradient iterations implemented with either ARG-matrix multiplication (see Note S3.3) or NumPy.

(F) CPU time required for testing a single trait or 50 traits in parallel on a simulated 10-kb window for sample sizes up to 500,000 individuals. Both ARG-RHE and GCTA-REML used random mutations generated from the ARG at a rate of $\mu = 10^{-6}$.

RESULTS

Scalability of ARG-matrix multiplication

We first evaluated the computational efficiency of ARG-matrix multiplication by comparing its runtime to that of two optimized numerical libraries, NumPy and SciPy, for simulated ARGs and genotype data covering a 5-Mb genomic region (STAR Methods). In these experiments, traditional matrix-based routines exhibited approximately linear scaling with sample size, whereas ARG-matrix multiplication scaled sub-linearly, achieving speedups of over $15\times$ at sample sizes approaching 500,000 individuals (Figures 1A and 1B). We observed that the performance benefit of ARG-matrix multiplication varies with the size of the analyzed region, with larger regions yielding greater speedups due to increased long-range LD among variants. In smaller regions (e.g., 10-kb windows; Figures 1C and S1), ARG-matrix multiplication still provided a 5–10 $\times$ speedup relative to traditional routines,

although the runtime scaled approximately linearly with sample size. Similar gains were observed in more complex numerical routines, such as the truncated eigendecomposition of the GRM used to compute test statistics within ARG-RHE. Using ARG-matrix multiplication to compute the top 20 principal components (PCs), we observed a $\sim 15\times$ speedup for 1,000,000 samples relative to an optimized implementation of the same randomized SVD algorithm in the PLINK2 software package⁴¹ (Figure 1D). Finally, we confirmed that the generality of ARG-matrix multiplication enables additional speedups in downstream analyses beyond those implemented in ARG-RHE. For example, in Figure 1E, we used ARG-matrix multiplication to compute a key quantity related to the best linear unbiased predictor (BLUP) of a trait by evaluating products involving the inverse GRM with the conjugate gradients algorithm from BOLT-LMM.⁴² This yielded an $\sim 9\times$ speedup for 200,000 individuals compared to a NumPy-based implementation. Additional details, including the use of this

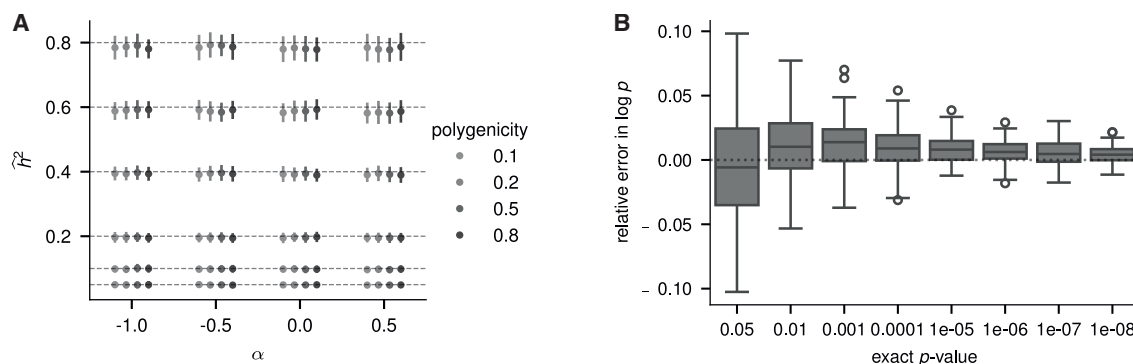


Figure 2. Accuracy of estimated heritability $\hat{h}^2$ and p values using ARG-RHE

(A) Estimated heritability $\hat{h}^2$ over a range of polygenicity levels, true h^2 (shown as dashed lines), and MAF-dependency parameter α . Heritability was estimated for a 10-Mb region and 20,000 samples, resampling mutations at a rate of $\mu = 10^{-8}$. Error bars show standard deviations estimated using 100 independent repetitions.

(B) Distribution of the relative error in estimated p values $(-\log_{10}p_{\text{approx}} + \log_{10}p_{\text{exact}})/(-\log_{10}p_{\text{exact}})$ corresponding to exact p value quantiles, using $L = 200$ leading eigenvalues across 100 random samples. p values were estimated for a 10-kb region and 5,000 samples, resampling mutations at a rate of $\mu = 10^{-6}$.

approach to perform scalable ARG-based LMM association, are provided in the [supplemental information](#).

Scalability, accuracy, and power of ARG-RHE

We next performed extensive simulations to verify the scalability, accuracy, p value calibration, and statistical power of variance component analyses using ARG-RHE. We first verified the scalability of ARG-RHE under large sample sizes by performing association in up to 500,000 individuals. We compared ARG-RHE to the approaches of Zhang et al.²⁰ and Link et al.,²³ which are based on first computing an ARG-GRM and then providing it to GCTA to estimate heritability using restricted maximum likelihood (GCTA-REML; [STAR Methods](#)). The results of these analyses are shown in [Figure 1F](#). As expected, GCTA-REML, which involves manipulating dense ARG-GRM matrices containing n^2 entries for n individuals, required substantially more computation and memory than ARG-RHE. As an example, testing an ARG region for association using 20,000 individuals with GCTA-REML required $\sim 300\times$ more time and $\sim 24\times$ more memory. The gap widened as the sample size increased, with an extrapolated difference of four orders of magnitude for an analysis involving 500,000 individuals. In addition, we verified that our extension of the RHE algorithm allows for efficiently estimating heritability across multiple traits in parallel, requiring negligible additional computation to analyze 50 traits compared to a single trait, as shown in [Figure 1F](#).

We evaluated the accuracy of using ARG-RHE to infer heritability across 96 genetic architectures simulated using different parameters (see [STAR Methods](#)). The results of these analyses are shown in [Figure 2](#) and [Tables S4](#) and [S5](#). Applied to 10-Mb gene windows, ARG-RHE yielded accurate estimates, with an estimated relative bias ranging from -3.6% to 1.0% , which is consistent with the results observed when applying the RHE algorithm to genotype data.³² On smaller 10-kb gene windows with simulated h^2 up to 0.02, ARG-RHE yielded slightly conservative estimates of heritability, with a relative bias ranging from -2.6% to -19.4% , depending on the α parameter, which gov-

erns the relationship between allele frequency and effect sizes.^{43,44} A small downward bias in ARG-based heritability estimates may also be observed if the mutation rate used when resampling mutations to work with Monte Carlo ARG-GRMs is not sufficiently large.²⁰

In addition to representing information on the genotypes of the analyzed samples, the ARG contains information on the age of the analyzed variants and can therefore be used to assess the phenotypic contribution of variants originating at different time-scales. These analyses could allow estimating the strength of natural selection on variants linked to the trait^{45–48} or could help quantify the relative phenotypic contribution of variation that is yet to be sequenced. To verify that ARG-RHE allows studying time-stratified variance components, we simulated phenotypic data using minor-allele-frequency (MAF)-dependent architectures under different values of α , corresponding to different effect size distributions for variants of different allele frequencies (see [STAR Methods](#)). Due to the link between the frequency and age of an allele,^{49,50} these simulated architectures also imply different effect-size distributions for variants of different ages, with larger negative values of α leading to larger effect sizes for rarer and more recent variants. These trends were captured by ARG-RHE when we applied it to jointly estimate multiple σ_g^2 values using implicit time-stratified GRMs (see [STAR Methods](#)), as shown in [Figure S5](#).

We next assessed the calibration of p values obtained using ARG-RHE when performing variance component association testing. We compared the p values from ARG-RHE, which are based on a fast randomized algorithm (see [STAR Methods](#)), to the exact p values obtained using Davies' method. To this end, we assessed the relative error between exact and approximate $\log p$ values, $(-\log_{10}p_{\text{approx}} + \log_{10}p_{\text{exact}})/(-\log_{10}p_{\text{exact}})$, as shown in [Figure 2B](#). Consistent with previous applications of this class of randomized approaches,³³ we found a small inflation in the mean estimated $\log p$ value, particularly at more extreme p values, which corresponds to the exact critical values being estimated to be positioned around 0.93 times their nominal

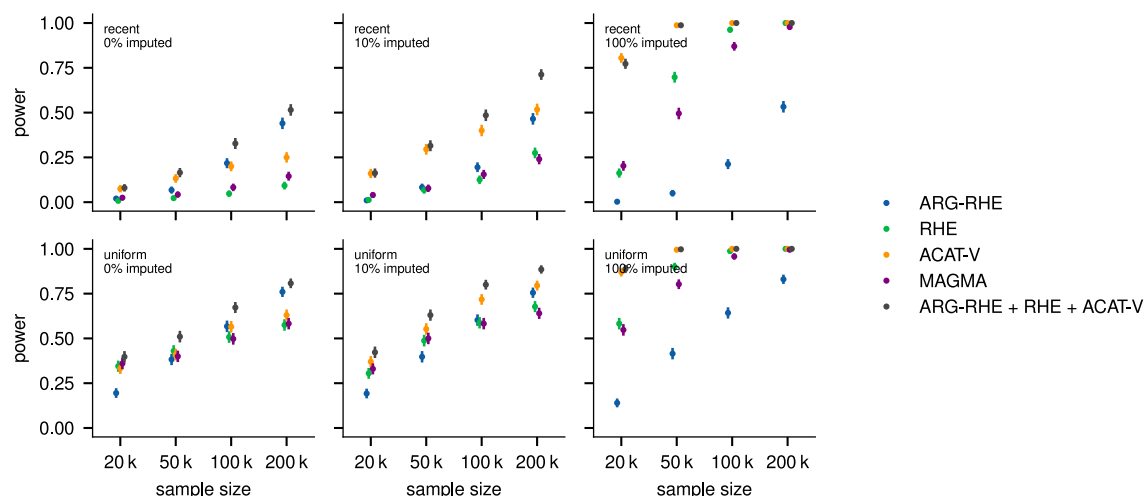


Figure 3. Statistical power of ARG-RHE with a varying proportion of causal variants included as imputation targets

We computed association power using a 10-kb gene window under polygenicity level $g = 0.05$ (fraction of causal variants among all sequencing variants simulated at a rate of 10^{-8}), local heritability $h^2 = 0.01$ (fraction of phenotypic variance explained by causal variants), and significance threshold $s = 2.5 \times 10^{-6}$. Causal variants are either sampled within the last 1,000 generations (recent) or uniformly throughout the entire genealogy (uniform), with 0%, 10%, and 100% of them being included as imputation targets. Error bars show standard errors estimated from 20 independently simulated ARGs each with 20 simulated traits.

tail quantile. To verify that this inflation is unlikely to meaningfully impact our UKBB analysis, we used the same set of inferred gene-based ARGs, simulated 50 non-heritable traits, and tested for association, observing 5 gene-trait pairs with p values that exceeded the Bonferroni-corrected threshold. This is a modest increase over the expected number of false positives under the null ($0.05 \times 50 = 2.5$, SD: ~ 1.9) and represents a negligible fraction of the total associations detected in real data, which are described below.

Next, we evaluated the ability of ARG-RHE, applied to SNP array data, to detect association signals complementary to those identified by other set-based methods, such as ACAT-V⁵¹ and MAGMA,⁵² when applied to imputed or sequenced variants. Results of these analyses are shown in Figure 3. To reflect varying genetic architectures and imputation accuracy, we simulated different scenarios by varying the age of causal variants and the extent to which these variants were polymorphic in the imputation reference panels (STAR Methods). In a scenario where 10% of causal variants were included in the imputation panel, comparable to recent estimates for the proportion of exome sequencing variants accurately imputed using the HRC panel,^{53,54} combining ARG-RHE with other tests yielded substantial power gains over imputation alone. The complementarity observed in these simulations was consistent with our analyses based on HRC-imputed UKBB data. As expected, in cases where all causal variants were accurately imputed, as in analyses based on directly sequenced data, ARG-RHE did not provide additional power. However, across a range of settings where imputation was incomplete, the combination of ARG-RHE and imputation-based methods consistently yielded the highest power, suggesting that ARG-RHE can effectively complement genotype imputation, particularly in scenarios where causal variants are poorly tagged by reference panels.

Finally, we assessed the extent to which using fast randomized algorithms results in a drop in association power compared to GCTA's likelihood-based approach. To this end, we compared ARG-RHE and GCTA-REML when applied to the same set of mutations resampled from the ARG, which are distinct from the underlying causal variants (see STAR Methods). Results for varying polygenicity and significance settings (see STAR Methods) are shown in Figure S6. At a significance threshold of 2.5×10^{-6} , ARG-RHE achieved a statistical power of around 60% relative to GCTA-REML for h^2 up to 0.02. Previous studies have observed a similar difference in statistical power between kernel-based score tests, such as in SKAT, compared to the likelihood ratio test used in GCTA.⁵⁵ However, due to its computational demands, GCTA cannot be applied to large-scale datasets for this analysis.

Overall, ARG-RHE provided substantial gains in computational speed and memory requirements compared to GCTA-REML while providing good accuracy and well-calibrated p values. The high computational scalability of this approach comes at the cost of a reduction in association power compared to exact likelihood-based strategies, which is consistent with previous applications of randomized approaches to variance component association testing. However, this decrease in power is offset by the possibility of applying ARG-RHE to datasets containing orders of magnitude more individuals.

Analysis of a UKBB ARG

We applied ARG-RHE to an ARG comprising 337,464 unrelated White British individuals from the UKBB dataset, which was inferred using 798,103 genotyped variants. We leveraged this ARG to test for an association between 21,159 protein-coding and non-coding RNA regions and 52 quantitative blood cell counts and serum biomarker level traits (see STAR Methods). We compared the results of ARG-RHE to those obtained by

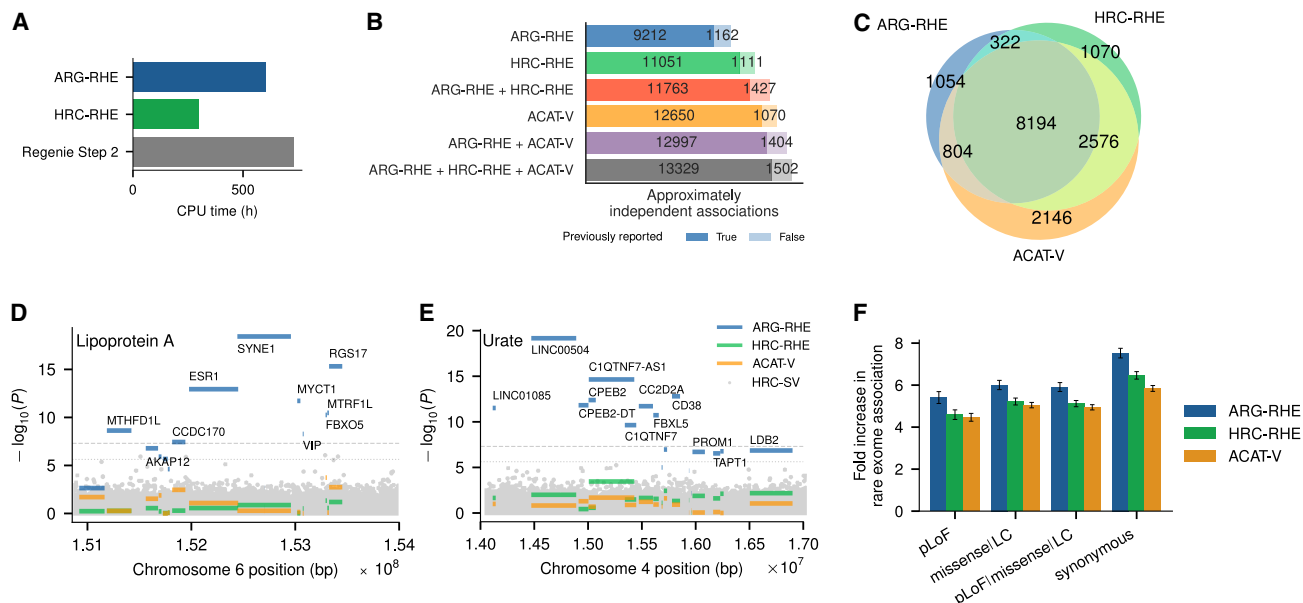


Figure 4. UK Biobank association analyses

(A) Total CPU time taken to test 52 traits for 337,464 unrelated White British individuals from the UK Biobank dataset. Both the ARG-RHE and HRC-RHE methods were run on a single thread, while Regenie was run using 12 cores; we report the total CPU time across all cores.

(B) Total number of LD blocks containing genome-wide significant gene-trait associations detected using different methods, including their combination using joint tests. We annotate the counts by whether the significant LD block-trait pair contains known gene-trait associations reported on the Open Targets platform (see [STAR Methods](#)).

(C) Overlap in LD blocks containing gene-trait associations detected using ARG-RHE applied to ARGs inferred from SNP array data or HRC-RHE and ACAT-V applied to HRC-imputed variants.

(D and E) Manhattan plot (D) for a genomic region containing genes, including SYNE1 and ESR1, associated with ARG-RHE to lipoprotein A levels and, similarly, (E) for a genomic region containing LINC00504 non-coding RNA, CPEB2, and CD38 genes, associated with serum urate levels. Dotted lines represent the genome-wide significance threshold for gene-based tests. Dashed lines represent the genome-wide significance threshold for HRC-SV single-variant tests.

(F) Concentration of significant exome gene-based associations with 52 blood traits after conditioning on the LD block having a genome-wide significant association in 337,464 unrelated White British individuals. Four annotation sets are used to filter the exome variants to be included in the gene-based SKAT-O test (see [STAR Methods](#); pLoF, high-confidence predicted loss-of-function variants; missense LC, low-confidence pLoF variants and in-frame insertions or deletions [indels]; missense LC and pLoF, the combination of the previous two masks; and synonymous). Error bars show the bootstrapped standard error in the estimated mean.

applying the same variance component association test to ~22 million variants imputed using the HRC reference panel (HRC-RHE). We also compared the results to associations detected by running Regenie to perform single-variant testing using the same set of HRC-imputed variants (HRC-SV) and using the ACAT-V algorithm to combine single-variant association statistics from imputed variants within gene regions (see [STAR Methods](#)). Results of these analyses are shown in [Figure 4](#). Applying ARG-RHE and HRC-RHE to this UKBB dataset required fewer computational resources compared to other efficient approaches for genome-wide association studies (GWASs). Applying HRC-RHE to test 52 traits across 21,159 regions spanning ~22 million imputed variants required 299.6 CPU hours, ARG-RHE applied to the UKBB ARG required 601.2 CPU hours, and applying Regenie to individually test the imputed variants required 730.8 CPU hours, without including step 1 of the analysis (see [Figure 4A](#)).

We first considered the sets of approximately independent genomic regions⁵⁶ in which gene-trait associations were detected using each method, as reported in [Figure 4B](#). Taken individually, running ARG-RHE, which relies on an ARG inferred using

only genotype array markers, identified 10,374 unique locus-trait associations, while HRC-RHE and ACAT-V, which rely on imputed data, identified 12,162 and 13,720, respectively. Consistent with recent analyses,^{20,23} however, the association signals that we detected using the ARG complemented the signals detected using imputation alone. Among the signals identified using ARG-RHE, 1,858 (18%) were not found by HRC-RHE, 1,376 (13%) were not found by ACAT-V, 1,770 (17%) were not found by HRC-SV, and 1,054 (10%) were not found by either HRC-RHE or ACAT-V ([Figures 4C](#) and [S7](#)). Performing a joint test using both imputed variants and the inferred ARG (see [STAR Methods](#)) yielded an increase in the number of associations. Combining ARG-RHE with HRC-RHE led to 13,190 associations, 8.5% more than when only using HRC-RHE; ARG-RHE with ACAT-V led to 14,401 associations, 5% more than ACAT-V alone; and combining all three tests led to 14,831 associations, 21.9% and 8.1% more than only using HRC-RHE and ACAT-V, respectively. The full list of all genome-wide significant associations found by ARG-RHE and estimated heritability $\hat{h}^2$ is provided in [Table S6](#).

We sought to validate the associations detected using ARG-RHE by looking for overlap with associations from other studies

reported in the Open Targets database⁵⁷ (see [STAR Methods](#)), which includes summary statistics from several single-variant and gene-based association studies, including analyses of large whole-exome or whole-genome sequencing datasets. Overall, we found 9,212 of the 10,374 regions associated with a trait by ARG-RHE to be reported on the Open Targets platform (see [STAR Methods](#)). These include established signals found using ARG-RHE but not detected in our analysis of this subset of UKBB samples using HRC-RHE, HRC-SV, or ACAT-V, such as the association between RNF168 and red blood cell count⁵⁸ and the association of SLC4A4 with neutrophil count.⁵⁹ Of the remaining 1,162 signals that are not present in Open Targets, several have been linked to related traits in other studies. Examples include associations found on chromosome 6 with lipoprotein A (Lp(a)), as shown in [Figure 4D](#). Among the highly significant genes detected using ARG-RHE, SYNE1 has a reported association with Lp(a) levels in the Hutterites, a founder population of South Dakota.⁶⁰ Associations with ESR1 have been reported in Klos et al.,⁶¹ and the RGS17 gene was significantly associated using exome sequencing data.⁵⁹ Other signals uniquely identified using ARG-RHE in our analyses and not reported in Open Targets include loci previously linked to traits that are related to those we have found to be associated. An example is the associations found on chromosome 4 with serum urate levels, as shown in [Figure 4E](#). The LINC00504 non-coding RNA and the CPEB2 gene were found to be associated with gout in a recent study involving ~2.6 million individuals,⁶² and the CD38 gene was found to be associated with gout in another study.⁶³

We also examined whether regions uniquely identified by ARG-RHE differ systematically from those uniquely identified by HRC-RHE or ACAT-V ([Table S7](#)). On average, genes with associations detected by ARG-RHE were shorter, exhibited higher recombination rates, and had lower imputation INFO scores. While some gene-based signals may reflect LD with neighboring regions, these trends support the hypothesis that ARG-RHE is particularly effective in regions with reduced imputation quality, where alternative methods may have reduced power. We further compared the associations detected by ARG-RHE and other methods to gene-based summary statistics derived from rare-variant tests in UKBB exome sequencing data.³⁵ [Figures 4F](#) and [S8](#) show the enrichment of exome-based signals across LD blocks found significantly associated with 52 blood phenotypes using each method. LD blocks containing signals detected by ARG-RHE using SNP-based ARGs were more likely to overlap with rare exome-based associations than those identified by HRC-RHE or ACAT-V, suggesting that ARG-RHE may be more sensitive to rare variant signals that are poorly imputed. Finally, using FUMA's GENE2FUNC pipeline,⁶⁴ we verified that genes associated with blood cell count traits by ARG-RHE showed enrichment for expression in whole blood in GTEx,⁶⁵ comparable to those identified by HRC-RHE and ACAT-V (see [supplemental information](#) and [Figure S9](#)).

Overall, the ARG-RHE algorithm was scalable enough to be applied to a genome-wide ARG inferred for 337,464 UKBB individuals using only genotyping array variants and allowed detecting associations that complemented those that would be identified using the same variance component test applied to HRC-imputed data alone. Signals uniquely detected using

ARG-RHE in these analyses were often implicated by studies that involved different cohorts or closely related traits. Further applications of ARG-RHE to non-European UKBB subgroups are described in Gunnarsson et al.⁶⁶

DISCUSSION

We developed methods to leverage ARGs to improve the efficiency of several genomic analyses, including LMM analyses of complex traits. Building on these advances, we introduce ARG-RHE, which combines moment-based heritability estimation³² with randomized algorithms for tail probability approximation,^{28,33} enabling scalable heritability estimation and variance component association testing with ARGs. In simulations, ARG-RHE was several orders of magnitude faster than the use of explicitly computed ARG-GRMs to estimate heritability²⁰ and to perform variance component association testing.²³ We verified the accuracy, calibration, and association power of ARG-RHE in simulations and confirmed its ability to identify complementary association signals, particularly when causal variants are poorly tagged by genotyped or imputed markers. Finally, we applied ARG-RHE to an ARG inferred for 337,464 UKBB individuals using only genotyping array variants, detecting associations that complement signals identified using the same gene-based variance component test on ~22 million HRC-imputed variants or single-variant LMM testing applied to ~54 million HRC-imputed variants.

Our study contributes to a growing body of work leveraging inferred genome-wide genealogies for the analysis of complex traits.^{20,22,23} One key advantage of using the ARG in this setting is that it provides a route to account for genomic variants that cannot be observed or accurately imputed. This is particularly the case in populations that are poorly represented in sequencing reference panels used for genotype imputation,^{20,23} as also evidenced by further applying ARG-RHE to non-European UKBB individuals in Gunnarsson et al.⁶⁶ Compared to testing individual edges of an inferred ARG, as done in Zhang et al.,²⁰ the use of a variance component test improves association power by pooling the effects of multiple variants and reducing the statistical burden by reducing the number of performed tests.²³ Our work substantially improves the computational efficiency of ARG-based variance component analyses of heritability and association, making this approach easy to apply to significantly larger samples.

Although our UKBB applications focused on association testing, our simulations show that ARG-RHE is scalable enough to infer genome-wide, rather than local, narrow-sense heritability. By leveraging information on allele ages contained in the ARG, it may also be used to gain insights into time-stratified phenotypic variance components. Natural selection acting on variants linked to polygenic traits often leads to larger per-allele effect sizes for variants of lower frequencies and younger ages,^{45–48} with stronger selection leading to a larger overall phenotypic contribution from rare and recent variation.^{44,67} ARG-based, time-stratified analysis may therefore quantify the strength of selection on variants across different time strata, complementing models that link selection and allele frequencies,^{43,44,67,68} as well as previous time-stratified approaches limited to higher-frequency variation.^{69,70} These analyses may provide insights into the fraction

of phenotypic variance explained by rare and recent alleles, which may not yet have been sequenced and are more likely to be population specific, with implications for study design and the generalizability of polygenic scores.⁷¹ However, we caution that obtaining unbiased heritability estimates is more challenging than testing for significance, as it imposes additional requirements on the accuracy of the inferred ARG. Current ARG benchmarks suggest that the accuracy of inferred topologies and branch lengths depends on factors such as marker density, recombination-to-mutation rates, and timescale,^{20,66} making robust time-stratified heritability analysis an interesting direction for future work.

Limitations of the study

We highlight several additional avenues for future development and limitations of this work. First, ARG-RHE currently only supports quantitative traits, but it could be extended to support the association of rare variants with binary traits that have low case/control ratios.^{72–74} Second, ARG-RHE implements a simple form of over-dispersion testing. Future work may involve implementing different strategies for weighting and pruning markers to improve power under various assumptions about the trait's architecture.^{51,52,75–78} Third, the multi-trait extension of the RHE-mc algorithm, which we use to reduce computational cost, relies on a single estimate of H and b , under the assumption that the intersection of individuals with non-missing phenotype data is large and that GRM estimation is not biased by missingness. This approach may be extended to obtain different H and b estimates for each trait. Fourth, as with other gene-based or single-variant association approaches, signals detected using ARG-RHE may reflect LD with variants in neighboring genes, and additional downstream analyses are required to more precisely localize causal variation. Finally, beyond heritability analysis, ARG-matrix multiplication can accelerate computations for other tasks, such as computing polygenic predictors (BLUP; Figure 1E) or performing mixed-model association (see supplemental information; also see Kalantzis⁷⁹). Extending these methods to additional analyses and applying them to real population-scale datasets is a promising direction for future work. Despite current limitations and areas of future work, the computational strategies developed here provide a highly scalable framework for leveraging inferred ARGs to complement genotype imputation in mixed-model analyses.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Pier Francesco Palamara (palamara@stats.ox.ac.uk).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- The code for ARG-RHE and other analyses described in this work can be downloaded at <https://github.com/PalamaraLab/ARG-LMM>.

- Additional information and software required to re-analyze the data reported in this paper are listed in the [key resources table](#) and are available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, P.F.P., J.Z., and G.K.; formal analysis, J.Z. and G.K.; funding acquisition, P.F.P.; investigation, J.Z. and G.K.; methodology, J.Z., G.K., and P.F.P.; project administration, P.F.P.; software, J.Z. and G.K.; supervision, P.F.P.; visualization, J.Z. and G.K.; writing – original draft, J.Z., G.K., and P.F.P.; writing – review & editing, J.Z., G.K., A.P., A.F.G., H.L., H.C., S.S., and P.F.P.

DECLARATION OF INTERESTS

A.F.G. is currently employed by deCODE genetics. H.C. receives consulting fees from Character Biosciences.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used Gemini 2.5 Pro to transcribe Java and C++ code into pseudocode for presentation. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- **KEY RESOURCES TABLE**
- **METHOD DETAILS**
 - Trait model
 - RHE algorithm
 - ARG-matrix multiplication and ARG-RHE
 - Additional mixed model analyses
 - Efficient ARG-based variance component association testing
 - Simulation experiments
 - Heritability estimation accuracy
 - Computational scalability
 - p Value calibration and statistical power
 - Analysis of UK Biobank data
- **QUANTIFICATION AND STATISTICAL ANALYSIS**

SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Deposited data		
UK Biobank genotype and imputation data	UK Biobank	https://www.ukbiobank.ac.uk/
Exome association summary statistics	Genebass	https://genebass.org/
Gene coordinates	HUGO Gene Nomenclature Committee	https://doi.org/10.1093/2Fnar/2Fgkaa980
LD blocks	LDetect	https://doi.org/10.1093/bioinformatics/btv546
GWAS summary statistics	OpenTargets	https://www.opentargets.org
Software and algorithms		
ARG-RHE	this work	https://github.com/PalamaraLab/ARG-LMM
arg-needle-lib	https://github.com/PalamaraLab/arg-needle-lib	https://doi.org/10.1038/s41588-023-01379-x
threads	https://github.com/PalamaraLab/threads	https://doi.org/10.1101/2024.08.31.610248
msprime	https://github.com/tskit-dev/msprime	https://doi.org/10.1093/genetics/iyab229
GCTA	https://yanglab.westlake.edu.cn/software/gcta/	https://doi.org/10.1016/j.ajhg.2010.11.011
Regenie	https://github.com/rcggithub/regenie	https://doi.org/10.1038/s41588-021-00870-7
PLINK2	www.cog-genomics.org/plink/2.0/	https://doi.org/10.1086/519795
ACAT-V	https://github.com/yaowuliu/ACAT	https://doi.org/10.1016/j.ajhg.2019.01.002
MAGMA	https://vu.data.surfsara.nl/index.php/s/lxDgt2dNdNr6DYt	https://doi.org/10.1371/journal.pcbi.1004219
FUMA, GENE2FUNC	https://fuma.ctglab.nl/	https://doi.org/10.1038/s41467-017-01261-5

METHOD DETAILS

In this section, we first introduce the trait model and notation used throughout the work. We then describe the Randomized Haseman–Elston regression (RHE) algorithm, on which ARG-RHE is built, along with its extension for parallel analysis of multiple traits. Next, we present an algorithm for fast multiplication between the genotype matrix and an arbitrary matrix, explain how it is used to implement ARG-RHE, and discuss additional applications, such as polygenic prediction and mixed model association testing. Finally, we describe the experimental setup used to benchmark these methods on simulated and real datasets.

Trait model

We assume a linear relationship between the phenotype and the genetic and environmental components,

$$y = W\gamma + X\beta + e,$$

where y is the phenotype vector for N samples, W denotes an $n \times c$ matrix of covariates, such as age and principal components (PCs) of ancestry, γ represents the corresponding fixed effects (including an intercept term), X is the $n \times m$ genotype matrix, β denotes the random effect sizes for each SNP, and e denotes random residuals so that $g = X\beta$ and e represent the random additive genetic and environmental components, respectively. We simplify the notation by assuming that the term used to represent covariates $W\gamma$ has been regressed out from the phenotype y and that y is standardized to have zero mean and unit variance. We assume that

$\beta \sim \mathcal{N}\left(0, \frac{\sigma_g^2}{m}I_n\right)$ and $e \sim \mathcal{N}(0, \sigma_e^2 I_n)$, where σ_g^2 and σ_e^2 capture the genetic and environmental variances and I_n is the $n \times n$ identity matrix.

We therefore have $g = X\beta \sim \mathcal{N}\left(0, \sigma_g^2 R\right)$, where $R = XX^T$ denotes the $n \times n$ genetic relatedness matrix (GRM). Assuming that g and e are uncorrelated, the sample variance-covariance is given by $V = \text{cov}(y) = \sigma_g^2 R + \sigma_e^2 I_n$. In addition, m should be viewed as a general scaling factor to ensure $\text{Tr}R = \text{Tr}I_n = n$, so that σ_g^2 and σ_e^2 are on the same scale. This coincides with the number of SNPs m under the assumptions that genotypes have been standardized and are in Hardy–Weinberg equilibrium. Under this setting, each SNP contributes equally to the overall phenotypic variance, which is equivalent to choosing $\alpha = -1$ in the model introduced in,⁴³ where the j -th variant with frequency f_j has effect size β_j with variance $\text{Var}(\beta_j) \propto [f_j(1 - f_j)]^\alpha$.

RHE algorithm

We describe the Randomized Haseman–Elston (RHE) regression algorithm,³¹ upon which ARG-RHE is based. The goal is to estimate narrow-sense heritability, the proportion of phenotypic variance attributable to additive genetic effects, defined as $h^2 = \sigma_g^2 / (\sigma_g^2 + \sigma_e^2)$.⁸⁰ Assuming a single genetic component, the phenotype is modeled as $y = g + e$, where $g \sim \mathcal{N}(0, \sigma_g^2 R)$, $e \sim \mathcal{N}(0, \sigma_e^2 I_n)$, and the GRM is defined as $R = XX^T/m$, with $X \in \mathbb{R}^{n \times m}$ the matrix of standardized genotypes. The moment-based estimators $\hat{\sigma}_g^2$ and $\hat{\sigma}_e^2$ are obtained by solving the system:

$$\begin{bmatrix} \text{Tr}R^2 & \text{Tr}R \\ \text{Tr}R & n \end{bmatrix} \begin{bmatrix} \hat{\sigma}_g^2 \\ \hat{\sigma}_e^2 \end{bmatrix} = \begin{bmatrix} y^T R y \\ y^T y \end{bmatrix}. \quad (\text{Equation 1})$$

To avoid forming R , RHE uses Hutchinson's stochastic trace estimator,⁴⁰ where traces such as $\text{Tr}R$ are approximated using B random vectors $z_1, \dots, z_B \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_n)$, as $\text{Tr}R \approx \frac{1}{B} \sum_{b=1}^B z_b^T R z_b$. As detailed in,³¹ the resulting computational cost is $\mathcal{O}(nmB)$, dominated by matrix-vector multiplications, and the memory cost is $\mathcal{O}(n + B)$, assuming a streaming implementation that avoids storing the GRM or the genotype matrix. In contrast, standard restricted maximum likelihood (REML) methods such as GCTA^{36,37} involve operations on dense $n \times n$ GRMs, which scale at least quadratically with N and require $\mathcal{O}(n^2)$ memory. An extension of this approach, RHE-mc,³² generalizes the model to multiple variance components.

We further extended this approach to estimate heritability across P phenotypes in parallel with minimal additional computational cost. This is achieved in three steps: (1) compute $y^T R y$ for each phenotype; (2) estimate $\text{Tr}R$ and $\text{Tr}R^2$ using shared random vectors; and (3) solve P linear systems using the same precomputed trace estimates. The overall complexity is dominated by terms proportional to $nm(B + P)$, which can be substantially more efficient than analyzing each phenotype independently, where the cost scales as $\mathcal{O}(nmBP)$. As discussed in the [supplemental information](#), additional care is required when phenotypes have differing patterns of missingness. Further implementation details, including extensions to the multi-component setting, are provided in the [supplemental information](#).

ARG-matrix multiplication and ARG-RHE

A wide range of statistical and population genetics analyses, including RHE and several algorithms for efficient mixed-model analysis, can be expressed in terms of multiplying the genotype matrix X by other matrices or vectors. When the mutations encoded in X are represented within an ARG, the structure of the graph can be leveraged to significantly reduce computational costs. This strategy, which we refer to as ARG-matrix multiplication, relies on graph traversal algorithms that efficiently propagate information across the nodes and edges of the ARG. We summarize the key components of this approach below and provide further details in the [supplemental information](#).

The ARG is a graph representing the genealogical history of a set of homologous genomes, where nodes correspond to the haplotypes of samples or their ancestors, edges represent the transmission of genetic material, and mutations are annotated along these edges. It can be efficiently represented in memory using graph-based data structures and traversed using algorithms that compute properties of its internal nodes, edges, and mutations. Such representations and algorithms were originally developed as part of genome simulators, including ARGON^{81,82} and msprime.^{83,84} These simulators use graph traversal to sample mutations in a memory-efficient manner,^{81,83} and similar strategies have been applied to compute ARG summaries such as identity-by-descent (IBD) haplotypes⁸¹ and other statistics.⁸⁵ More recently, related methods have been designed to accelerate computation on other genomic graph structures.⁸⁶ As detailed in the [supplemental information](#), these algorithms leverage memoization, allowing properties of nodes or mutations, such as the set or number of descendants or carriers, to be computed efficiently by reusing results from nearby nodes or mutations.

Our approach for ARG-matrix multiplication builds on these ideas to efficiently compute matrix products of the form $U \cdot X$ (left multiplication) and $X \cdot U$ (right multiplication), where X is the $n \times p$ genotype matrix for n individuals and p mutations in the ARG, and U is an input matrix of compatible dimensions. Due to linkage disequilibrium (LD), the set of carriers of one mutation often overlaps significantly with carriers of other nearby mutations. In such cases, matrix-vector products can be computed by reusing results from related mutation vectors, enabling computations that scale sublinearly with respect to the number of samples, depending on the extent of LD present in the data.

We implemented the RHE algorithm from the previous section (RHE algorithm) using this approach, which we refer to as ARG-RHE. ARG-RHE leverages ARG-matrix multiplication routines to efficiently perform numerical computations involving repeated matrix-vector products, such as Hutchinson's trace estimator and computing the leading eigenspace of the GRM $R \propto XX^T$ for significance testing (see previous section). ARG-RHE, as well as other ARG-based analyses that can be built on this framework, can be applied either to the existing mutations present on an inferred ARG or to newly resampled mutations, as described in.²⁰ When newly resampled mutations are used, ARG-RHE captures the effects of variants likely present at the sequencing level but not directly observed in the data used to infer the ARG. This provides a computationally efficient alternative to the approach implemented in,²⁰ where mutations were randomly resampled from an inferred ARG, written to disk, and explicitly used to construct

ARG-GRMs for downstream analysis with GCTA. Such explicit construction limited those analyses to relatively small simulated samples.

Additional mixed model analyses

Zhang et al.²⁰ leveraged the ARG to perform additional LMM analyses, such as polygenic prediction and mixed-model association. However, due to computational constraints, in their UK Biobank analyses they used BOLT-LMM to implicitly construct GRMs from common SNP array variants, applying the ARG only to test for association with unobserved variation rather than to construct an ARG-GRM. While ARG-RHE and our large-scale applications focus on heritability estimation, ARG-matrix multiplication can also accelerate other ARG-based LMM analyses. For example, polygenic prediction and association testing require solving linear systems of the form $Vx = y$ using conjugate gradients,⁴² where $V = \sigma_g^2 R + \sigma_e^2 I_n$. In the [supplemental information](#), we describe an approach for ARG-based polygenic prediction and association testing that leverages ARG-matrix multiplication (also see Kalantzis⁷⁹).

Efficient ARG-based variance component association testing

We extend ARG-RHE so that, in addition to obtaining an estimate of the variance explained by mutations that may be generated using the ARG within a region, we can also efficiently assess whether this estimate is statistically significant, thereby performing variance component association testing.

For ARG-based association testing in this paper, we focus on the case of a single genetic variance component and a phenotype y standardized to unit variance, solving [Equation 1](#) while maintaining that $\text{Tr}R = n$ leads to a closed-form expression for the heritability estimator

$$\hat{\sigma}_g^2 = \frac{y^\top (R - I_n) y}{\text{Tr}R^2 - n}. \quad (\text{Equation 2})$$

We consider the case where R is a local GRM, constructed by resampling mutations within the ARG in a region of interest. To test for association with the phenotype y , we focus on the distribution of the estimator $\hat{\sigma}_g^2$ under the null hypothesis $H_0: \sigma_g^2 = 0$, where $y \sim \mathcal{N}(0, I_n)$. The random part in the numerator of [Equation 2](#) gives the test statistic $T = y^\top R y$. After eigen-decomposing $R = \Sigma^\top \Lambda \Sigma$, where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ holds the eigenvalues of R in decreasing order and columns of Σ store the corresponding eigenvectors, we have

$$\Pr(T < t | H_0) = \Pr(y^\top R y < t | H_0) = \Pr((y^\top \Sigma^\top) \Lambda (\Sigma y) < t | H_0) = \Pr\left(\sum_{i=1}^n \lambda_i \chi_{1,i}^2 < t\right), \quad (\text{Equation 3})$$

where $y \sim \mathcal{N}(0, I_n)$ is invariant under the action of Σ , and $\chi_{1,i}^2$ are i.i.d. chi-squared variables with 1 degree of freedom. This test statistic coincides with the score-based test adopted in methods such as the sequence kernel association test (SKAT),²⁸ which is instead derived by considering the derivative of the restricted likelihood at $\sigma_g^2 = 0$, although SKAT has a different default weighting of variants based on their allele frequencies. We note that, while it is also possible to efficiently estimate the standard error of $\hat{\sigma}_g^2$,³¹ using a Wald test on H_0 requires T to be asymptotically normally distributed. However, since T in [Equation 3](#) has significant skew and kurtosis, convergence to a normal distribution depends on the number of non-zero λ_i . This is bounded by the number of variants in a gene region and is usually not sufficient to apply large-sample asymptotics.⁸⁷ Hence, to obtain a tail probability that is precise enough for association testing but also computationally efficient, we follow the strategy recently proposed in fastSKAT,³³ which relies on a mixture of moment-based and saddle-point approximations. More precisely, we first use ARG-matrix multiplication within a randomized SVD algorithm^{38,39} (also adopted to compute principal components and eigenvalues in PLINK2⁴¹) to estimate $\lambda_1, \dots, \lambda_L$ as the top- L eigenvalues of Λ . We then approximate the right-hand side of [Equation 3](#) using a truncated sum of $L+1$ chi-squared variables, which take the form

$$\tilde{T} = \sum_{i=1}^L \lambda_i \chi_{1,i}^2 + a \chi_d^2, \quad (\text{Equation 4})$$

where

$$a = \frac{\sum_{i=L+1}^n \lambda_i^2}{\sum_{i=L+1}^n \lambda_i} = \frac{\text{Tr}R^2 - \sum_{i=1}^L \lambda_i^2}{\text{Tr}R - \sum_{i=1}^L \lambda_i},$$

$$d = \frac{(\sum_{i=L+1}^n \lambda_i)^2}{\sum_{i=L+1}^n \lambda_i^2} = \frac{(\text{Tr}R - \sum_{i=1}^L \lambda_i)^2}{\text{Tr}R^2 - \sum_{i=1}^L \lambda_i^2},$$

are scalars that satisfy the moment-matching conditions under the null $\mathbb{E}_{H_0}[T] = \mathbb{E}_{H_0}[\tilde{T}]$ and $\mathbb{E}_{H_0}[T^2] = \mathbb{E}_{H_0}[\tilde{T}^2]$. We estimate these quantities using the leading eigenvalues and Hutchinson's estimator for the trace of powers of the GRM R . Next, we estimate association p -values via a less accurate but fast moment-based estimation approach.⁸⁸ If the p -value is below a specified threshold, which we set to 0.01, we proceed to estimate the tail probability more accurately through a saddle-point approximation technique.⁸⁹ This approach is designed to avoid the numerical instability of saddle-point approximation near the mean of the distribution. As done

in the computation of point estimates for heritability, we use ARG-matrix multiplication to directly perform these computations within the ARG.

Simulation experiments

We evaluated ARG-RHE by testing its scalability, the accuracy of its h^2 and p -value estimates, and its power when applied to perform variance component association analysis.

To work with Monte Carlo ARG-GRMs (i.e., ARG-GRMs estimated using mutations randomly resampled from the inferred ARG at high rate, see²⁰), we discarded the original mutations used to infer the ARG and resampled a new set at a rate of 10^{-6} for 10-kb windows and 10^{-8} for 10-Mb windows. This yielded a genotype matrix X , which we zero centered and normalized by $[f(1-f)]^{1/2}$, where f denotes the allele frequency. With this approach, the ARG-GRM can be written as $R = XX^T/m$, where m is a normalizing factor such that $\text{Tr}R = n$, and used to calculate p -values of the test statistic $T = y^T R y$. In experiments involving GCTA, we explicitly formed the ARG-GRM after writing these mutations to disk, as done in.²⁰ When running the ARG-RHE algorithm, the mutations remained stored in the ARG and were used to obtain heritability estimates and assess significance. We used 200 random vectors to compute Hutchinson's trace estimator and the top 200 leading eigenvalues obtained from the randomized SVD. All tests were performed using Intel(R) Xeon(R) Gold 6240R 2.40GHz CPUs.

Heritability estimation accuracy

To assess the accuracy of the heritability estimates, we generated ARGs containing 20 000 individuals under a constant effective population size $N_e = 50\,000$ and recombination rate $\rho = 10^{-8}$, spanning 10 kb and 10 Mb gene windows. We then generated heritable traits with h^2 ranging from 0 to 0.8. We varied polygenicity from 10% to 80% and the MAF-dependency parameter α from -1 to 0.5 (Table S4). We generated 100 independent simulations for each combination of parameters. When estimating h^2 using ARG-RHE, we assumed the correct α used to generate the phenotypes.

We also performed experiments in which we estimated σ_g^2 using time-stratified GRMs. For these analyses, we simulated 10-Mb windows on 20 000 individuals as above. We additionally partitioned the resampled genotypes into 6 GRMs containing an equal number of mutations based on their age quantiles, and jointly estimated the σ_g^2 across time bins using multi-component ARG-RHE. We generated heritable traits with $h^2 = 0.4$ and effect sizes α ranging from -1 to -0.25 , and formed the ARG-GRMs using a fixed $\alpha = -1$.

Computational scalability

We simulated 10-kb and 5-Mb regions with up to 10 000 000 diploid individuals under a demographic model inferred from the GBR population in the 1,000 Genomes Project dataset.⁹⁰ These simulations were used to assess the scalability of the ARG-matrix multiplication routine and its applications, using resampled mutations generated at specified rates. We compared the total CPU time required with that of other baselines, including NumPy and PLINK2, applied to the same set of resampled mutations. To assess computational efficiency, we again simulated 10-kb gene regions with up to 500,000 individuals and compared the costs of running ARG-RHE to the use of GCTA to estimate heritability with an explicitly constructed ARG-GRM, as done in Zhang et al.²⁰ and Link et al.²³ When comparing the use of GCTA to the ARG-RHE algorithm, we used the same set of resampled mutations to explicitly construct the ARG-GRM using GCTA (v1.94.1) or to run the ARG-RHE algorithm, which is implemented in the arg-needle-lib library (see URLs). We measured the computational time required to run GCTA on pre-computed ARG-GRMs, excluding the time required to construct such ARG-GRMs, and compared with the time required to compute heritability and test for significance using ARG-RHE. Unlike ARG-RHE, GCTA does not support analyzing multiple phenotypes in parallel, so in these benchmarks, we tested ARG-RHE with a single phenotype, as well as with 50 random phenotypes processed in parallel.

p Value calibration and statistical power

We performed additional simulations to verify that variance component testing using ARG-RHE results in sufficiently calibrated p -values. To this end, we obtained exact quantiles for the test statistic T at p -values ranging from 0.05 to 10^{-8} for simulations consisting of 5,000 individuals. We used Davies' method,⁹¹ obtaining a complete SVD of R to verify that the randomized approach used by ARG-RHE produces accurate p -values for a range of significance levels. For each quantile, we tested 20 independently generated ARGs each with 5 sets of random trace estimator seeds, totalling 100 estimators at each p -value threshold. We also verified the empirical null distribution of the p -values by testing 50 randomly generated non-heritable traits against the same inferred gene-based ARGs from 3 37 464 UK Biobank individuals used in our real data analysis.

We evaluated the statistical power of ARG-RHE under a range of setups to assess the complementarity of its gene-based association signals to alternative approaches such as ACAT-V⁴⁹ and MAGMA⁵² applied to imputed or exome-sequenced data. We simulated ARGs covering a 20-Mb region under a demographic model inferred from the GBR population in the 1,000 Genomes Project dataset,⁹⁰ including up to 2 000 000 diploid individuals, along with an additional 10% used as an imputation reference panel. Sequencing variants were simulated by sampling mutations from the true ARG at a rate $\mu = 10^{-8}$. For samples not in the reference panel, we filtered mutations to match the allele frequency spectrum of UK Biobank SNP array data, following the approach in.²⁰ These SNP variants were also used as input to Beagle v5.5 for imputation and to Threads v0.1⁶⁶ to construct the inferred ARG, using default parameters.

We generated synthetic phenotypes by selecting the central 10-kb segment of the ARG as the gene region and randomly choosing causal variants from the simulated sequencing variants within this region, sampling their effects as $\beta \sim \mathcal{N}(0, [f(1-f)]^\alpha)$ with $\alpha = -1$. We added additional noise to achieve a trait heritability of $h^2 = 0.01$. To avoid simulating rare variants with unrealistically large effect sizes, we restricted causal variants to those with minor allele count (MAC) ≥ 5 . To simulate causal variants that are not included in the imputation panel, we generated mutations on the same ARG using a different random seed. To model different causal allele frequency patterns, we sampled causal variants either from recent time (i.e., up to 1,000 generations) or uniformly across the entire ARG. After evaluating the effect of varying the resampling rate on ARG-RHE's power (see [Figure S2](#)), we chose 10^{-6} as the default mutation sampling rate on inferred ARGs. For each parameter combination, we generated 20 sets of ARGs and sequencing variants, each with 20 independently drawn sets of causal mutations and phenotypes. Statistical power was assessed at $p = 2.5 \times 10^{-6}$, corresponding to a Bonferroni-corrected threshold assuming 20 000 gene regions and a family-wise error rate of 0.05.

We also compared the statistical power of ARG-RHE testing, which uses a moment-based approach, to that of GCTA, which relies on a more accurate but less scalable likelihood-based approach. We ran GCTA using two sets of variants for the GRM. In one case, we computed the local GRM using the actual set of sequencing variants used to simulate the phenotype. This simulation provides an upper bound to the statistical power that can be achieved using ARG-RHE, which relies on a different set of resampled mutations. In the other case, we explicitly constructed a Monte Carlo ARG-GRM, using the same set of resampled mutations used by ARG-RHE as input GRM for GCTA to test for association. This simulation is meant to assess the extent to which the approximate randomized approach of ARG-RHE results in decreased association power compared to GCTA's likelihood-based approach. We again assessed the statistical power at $p = 2.5 \times 10^{-6}$, and performed 100 repetitions for each combination of parameters.

Analysis of UK Biobank data

We applied ARG-RHE to perform variance component association testing using an ARG inferred for 337,464 unrelated white British individuals from the UK Biobank dataset,² using a single variance component for resampled variants within a gene-based window. The ARG was inferred using *Threads*⁶⁶ (v0.1) applied with default parameters to 798,103 genotyped variants, each with less than 10% missing values, which were computationally phased using *Beagle* (v5.1).⁹² We tested for association between 21,159 protein-coding and non-coding RNA regions from the HUGO Gene Nomenclature Committee (HGNC) table on the University of California, Santa Cruz (UCSC) genome browser (see URLs) and 52 quantitative blood cell traits, including 27 blood cell indices and 25 blood biochemistry marker levels. The lists of regions and traits tested are provided in [Tables S1](#) and [S2](#).

We compared the results obtained using ARG-RHE to those obtained by applying the same variance component association test of [Equation 4](#) to genotype dosages imputed from the Haplotype Reference Consortium (HRC) reference panel, as performed in.² We included HRC-imputed variants with MAC ≥ 5 , INFO score ≥ 0.3 , missingness $\leq 10\%$, and Hardy-Weinberg equilibrium $p \geq 10^{-15}$, leading to a total of 54,356,141 variants tested for standard single-variant association testing using *Regenie*.⁷³ We refer to these single-variant analyses of HRC-imputed data as HRC-SV. We also computed the test statistic of [Equation 4](#) using 22,460,173 HRC-imputed variants that overlap a gene region, which we refer to as HRC-RHE. Finally, we also performed ACAT-V gene-based testing⁵¹ by combining HRC-SV p -values from a gene region using uniform weights. We set the genome-wide significance threshold to be $0.05/21159 \approx 2.36 \times 10^{-6}$ for gene-based tests, and 10^{-9} for single-variant tests.

We compared the signals from both gene-based and single-variant tests by grouping p -values from the tested regions into approximately independent LD blocks, using *LDetect*.⁵⁶ We then checked whether each block-trait pair contains genome-wide significant signals. We also considered a simple joint test that combines ARG-RHE and HRC-RHE: for each gene region, we took the smaller p -value $p_{\min}$ from the two methods and treated them as independent tests by comparing $2p_{\min}$ to the same genome-wide significant threshold. We used a similar method for combining ARG-RHE, HRC-RHE, and ACAT-V results. When running ARG-RHE, we used $L = 200$ leading eigenvalues in estimating the tail distribution and $B = 200$ random vectors in Hutchinson's trace estimator.

To preprocess the analyzed traits, we followed the approach of.⁵⁹ We first stratified samples based on sex and menopause status and applied a rank-inverse-normal transformation (RINT) to the adjusted phenotype values. We regressed out covariates, which included age, age squared, alcohol use, smoking status, height, body mass index, assessment center, genotyping array, and 20 principal components. We then applied RINT a second time and merged all strata. Missing phenotypes were mean-imputed as 0. To verify that the top 20 PCs can account for most of the population structure, we performed simulations in which heritable traits were simulated using HRC-imputed variants on odd chromosomes and residualized using the leading 20 PCs. We then tested for associations with regions on the even chromosomes and observed no significant deviation from the null (see [Figure S3](#)). We also verified that mean imputation at 0 does not inflate the type I error rate (see [Figure S4](#)).

We verified the detected associations by checking whether they were previously reported on the Open Targets platform (version 22.10).^{93,94} Specifically, for each significant gene-trait association, we used Open Targets' locus-to-gene pipeline to check for associated variants from previous studies involving the same gene. We considered an LD block-trait pair as previously detected if the LD block contained at least one such associated gene for the trait in Open Targets. For this analysis, we matched our 52 quantitative traits to 495 traits with closely related descriptions available on Open Targets ([Table S3](#)).

We additionally validated our genome-wide significant associations against signals detected using *Genebase*³⁵ a genome-wide association study based on exome sequencing data from a larger UK Biobank cohort. For each gene-trait pair, we obtained summary statistics under four variant masks: synonymous, high-confidence predicted loss-of-function (pLoF), low-confidence pLoF and in-frame indels (missense LC), and the combined missense LC & pLoF variants. Each mask was analyzed with *SAIGE-GENE*⁷⁷ under

three association methods: burden, SKAT, and SKAT-O. We grouped these gene-based exome summary statistics into our approximately independent LD blocks. To quantify enrichment, we computed the fold-increase in probability that an LD block contains a genome-wide significant exome signal, conditional on it being significantly associated by one of the tested methods (ARG-RHE, HRC-RHE, or ACAT-V). We compared this against the baseline probability of association, calculated across all LD blocks and irrespective of their significance under any method. We estimated the standard errors by resampling LD block-trait pairs with replacement. Finally, we focused on the subset of 27 blood cell count traits, and evaluated the detected gene associations using the GENE2FUNC module from FUMA⁶⁴ to annotate their expression levels in whole blood (see [supplemental information](#)).

QUANTIFICATION AND STATISTICAL ANALYSIS

All statistical analyses were performed using the following software: Python 3.11.3, SciPy 1.11.4, NumPy 1.26.4, chiscore 0.2.3, GCTA 1.94.1, MAGMA v1.10, and regenie 3.4.1. All methodological details can be found in the [method details](#), and all statistical tests are named as they are used.